

Virtual Reality Based Human-Computer Interaction System for Metaverse

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ABSTRACT

This work aims to evaluate the performance and effect of a Human-Computer Interaction (HCI) system in the Metaverse with real-time interaction and immersion. This work introduces the Metaverse with HCI functions of 3D face reconstruction, handshake and hug, multi-user synchronous editing environment, all information is stored in the blockchain. The performance of the HCI functions are also evaluated. The results demonstrate that the overall correct rate of the subjects by the HCI system is 95.45%. Users' social acceptance of VR interactive systems, action interaction usability, user emotions, and user satisfaction all show high scores. This work contributes to the performance evaluation and optimization of the HCI systems in the Metaverse.

Keywords: Metaverse, Human-Computer Interaction, 3D Interaction, Virtual Reality, performance evaluation.

1 INTRODUCTION

With the development of science and technology, Virtual Reality (VR) technology has been dramatically assisted. The proposal of the Metaverse concept presents a new horizon for the interaction between the virtual world and the natural world [1]. Research scholars have made a lot of efforts on the immersive experience of the Metaverse and built various HCI systems. Liu et al. (2022) [2] explored the reliability of intelligent computer-implemented communication and established an HCI-based visual feedback system for augmented and alternative communication. The research and analysis of the above scholars suggest that common network social applications have been unable to meet people's needs, and the emergence of the Metaverse makes it possible for users in different regional spaces to have physical interaction.

In conclusion, improving the performance of the HCI system in the Metaverse has extremely practical significance for the development of intelligence in the social field. This work innovatively proposes a three-dimensional (3D) face reconstruction method based on weakly supervised learning (WSL) to build an HCI system with a realistic interactive immersion in the Metaverse. In addition, the system implements and optimizes the hugging and handshake of the players. Moreover, multi-user synchronous editing environment is implemented. The most important under the background of web3, all information is stored in the blockchain. Finally, the system has been evaluated from several perspectives with good results. This work can provide new research ideas for the study of VR immersion in the Metaverse.

2 RESEARCH METHODS

The internal functional modules of the HCI system mainly include a global clock module, an image data acquisition module, a control module, a human motion tracking algorithm module, an

attitude data acquisition module, an attitude calculation module, and a data transmission module. They cooperate with peripheral circuits to realize various functional requirements of the system. A D face reconstruction algorithm based on WSL is proposed in the image acquisition module. It can construct a 3D face from 2D photos in the virtual environment. In the human motion tracking algorithm module, the virtual space of Metaverse is used to interact according to the different locations of users. In this way, users in different geographic spaces can communicate and exchange information to complete the same collaborative task similar to the real scene, such as shaking hands, and hugging. Multiple users can synchronous edit environment, for example build a building together like in the real world. In the data transmission module, the use of blockchain technology to store text and voice can effectively save the user's historical data. First, the local system is used to enter the blockchain. Then, the text content is entered and connected to the written smart contract through Infura. Besides, the update method of the smart contract is invoked to write the data to the blockchain for permanent storage.

3 EVALUATION

Experiments are carried out on the VR scene of the Metaverse HCI system established above to verify the performance of the above system in all aspects. In this experiment, the user's information is obtained by building an experimental system; the accurate real-time parameters are obtained with the help of the 3D face reconstruction method in the intelligent HCI system and the human body posture detection method. At the same time, it cooperates with the host computer software to test the HCI function of the design system to verify the reliability of the design system in solving related problems.

Figure 1 illustrates the performance and real-time evaluation of the face reconstruction algorithm for the HCI system of the Metaverse.

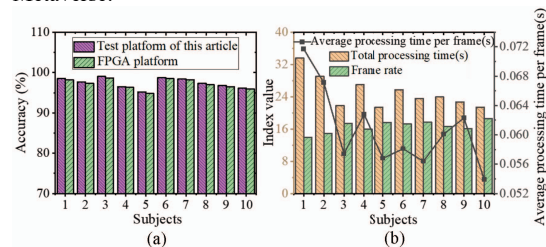


Figure 1: Face reconstruction and real-time evaluation results of the HCI System of the Metaverse (a. evaluation of 3D face reconstruction algorithms; b. assessment of real-time evaluation of the system)

Figure 1a suggests that the accuracy of the 3D face reconstruction algorithm proposed here can reach about 95% under the same platform, although there are some differences among different individuals. However, the accuracy of the same individual varies from platform to platform. This is because the data on the field bus is of the default integer type to reduce the consumption of resources in the chip and the amount of algorithm calculation, losing some accuracy in the calculation process.

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It can be seen from Figure 1b that with the increase of the input data sample size, the overall frame rate presents an upward trend. On the contrary, the average processing time of each frame of image data presents a downward trend. The result means that the larger the input data sample size, the more image data the algorithm in the system can further improve its analysis speed, which reflects that the algorithm in the system has the ability to learn independently. Finally, it is found that the test platform based on PC in this experimental system can reach the processing speed of 18 frames of image data at the highest. It can meet the real-time processing requirements of the system output image.

Figure 2 shows the evaluation results of the control function of the head and body limb movements of the HCI system of the Metaverse.

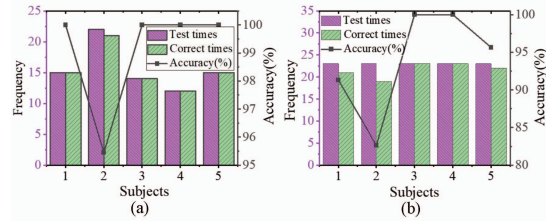


Figure 2: System control module performance evaluation results (a. head control accuracy evaluation; b. accuracy assessment of body and limb movements)

Figure 2a suggests that subject No. 2 makes a judgment error during the experiment, and the overall correct rate is 95.45%. The correct rate of several other subjects is 100%. The average correct rate of 5 subjects can reach 99.09%.

As can be seen from Figure 2b, except for subject No. 2, the correct rate of the rest of the subjects attains more than 90%, of which two subjects reached 100%. The overall average correct rate reaches 93.91%, indicating that the system has a high degree of reliability.

In a word, the experimental environment shows that the algorithms used in this system have high accuracy and meet the real-time requirements. Finally, the test of the HCI function proves that the system has high reliability.

Figure 3 and Figure 4 present the analysis of social satisfaction and user experience scores from the perspective of users

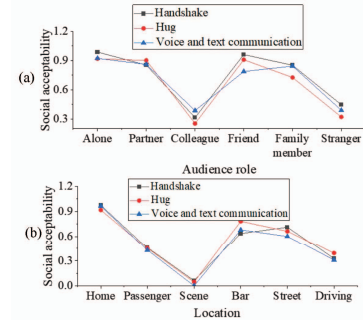


Figure 3: Survey results of users' social satisfaction with different functions of the system (a. users with different roles; b. Users in different locations)

Figures 3 are social acceptance curves for hugs and handshakes and social acceptance curves for speech-to-text communication. It can be seen that the average acceptance rating in all contexts is above 0.5. The VR-based HCI system reported here has a significant impact on social acceptance at the 0.5 level. In addition, user hugs and handshakes were rated higher in social acceptance than the functional curve for communicating via voice and text.

This result shows that the behavior of hugging and shaking hands of users in the virtual reality system is more readily accepted by society. It also reflects from the side that shaking hands and hugging between users can help them become more familiar and accepted with each other more quickly.

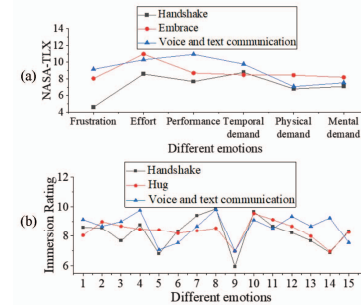


Figure 4: User emotion and immersion ratings (a. emotion ratings; b. immersion rating)

As can be seen from Figure 4a, in the process of experiencing the system, the proportion of positive emotional states is high, and the proportion of negative emotions is low. For example, the proportion of users who express frustration is the lowest. This result reflects the high availability of the system. Figure 4b suggests that users generally rate immersive experiences highly. The comprehensive scoring curve of the three functional modules (handshake, hugging, and voice-text interaction) demonstrates that the immersion experience scores are all above 6 points, and the lowest is close to 6 points. This suggests that when participants are placed in the virtual environment established here, the sensory system processes visual and other perceptual data from the virtual world in a manner very close to that in the natural environment. This indicates that users have a sufficient immersive experience in the process of the HCI VR system established here.

4 CONCLUSION

VR in the Metaverse is more of a subjective digital experience than a cold concept of territory. This work evaluates the performance of a Metaverse HCI system with realistic interactive immersion with multiple modules from different perspectives. The results demonstrate that the system has high scores in social acceptance, action interaction usability, user emotion, and user satisfaction. The shortcoming of the research is that complex and intelligent interaction between users in the virtual space is ignored, which is worth considering in further research.

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